

having these AI models talk to each other, instead of re-reading lots of data over and over. Like when you learn a new thing, you don't re-read all the stuff you've ever learned, right? You keep updating yourself. For example, when you add a word to a language model, it's one word. You don't add all the words you've ever learned. That's a really interesting dynamic.

What are your thoughts on open source software and hardware, especially in AI systems?

Intel, when they built their CPU, became the open hardware standard because they did a very good job of documenting, exposing their instruction set, and providing tools so everyone could use it. Way back, Intel architecture was built by seven different manufacturers. People were willing to write Assembly language programs for that.

Now on GPUs, the low-level instruction set is actually somewhat difficult to use, and the GPU vendors provide all the compiler software. You can write code at a high level and then compile it through the hardware. The GPU vendors actually change their instruction set almost every generation, so the user never sees the hardware directly. Tenstorrent is building the hardware and the software stack both. Now, we're going to open-source that software stack. So, people, if they want to, can go to the hardware level.

Could you elaborate on a few concepts that have come up recently—like Software 2.0 and brain-like execution?

The big idea is, in Software 1.0 people write programs to do things. For 2.0, people use data to train models. For example, you can train a chess program with a billion chess moves. Or you can build a model of chess and a simulator, and then have the simulator compete with itself and slowly learn

what the good moves are.

Where do you get the data for Software 2.0? You could get data from the simulation, from scraping the internet, etc. The data could be images, text, or scientific equations. At the hardware level, we don't really care about where the data comes from. Pretty much no matter what, it turns into these graphs of computations.

You don't want to fill the whole GPU with one big computation. But the way the models on GPUs are written, they essentially do the whole thing. Even in executing AI graphs, you go through the whole graph, no matter what. That's not how your brain works; your brain does lots of small computations. If you're thinking about animals, it fires up one part of your brain. If you're thinking about a book, it fires up a different part. That's called conditional execution.

Do you see AI computing becoming more accessible to the general public than before?

Consumer products are very successful when they're under a thousand dollars right now. Years ago, when an American consumer went to a store to buy a TV, and if it was under \$500, he could just buy it. If it was over \$500, they would go home and research it first and figure out which one to buy. Right now, due to inflation, that number is about \$1200.

AI computers get expensive very fast. You can end up spending \$1000 over a weekend running some models! Many startups are working to make this affordable. For example, Tenstorrent's list price plan for AI processing is to be about 5 to 10 times cheaper than the current market rate. We think that makes it more accessible. On the software side, if we can say we have a model compiling and running easily without requiring five IT people for support, it is more accessible. **EFY**

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-Jeff Bezos

